

5 FORCE

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1-D motion x vs t is completely artificial

Reality is force, Push/Pull. This goes

back to Isaac Newton 1687 Principia.

source of physics and calculus

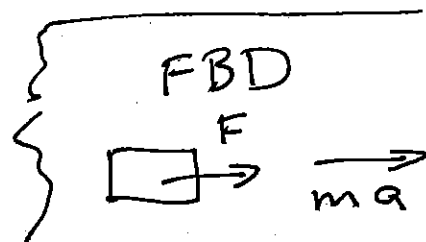
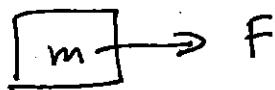
Unit of Force is Newton $1N = 1 \frac{kgm}{s^2}$

N1 Obj. in motion remains in motion;
obj. at rest remains at rest.

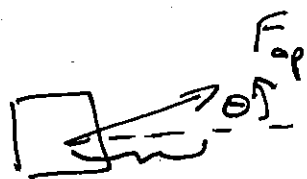
N2 $F = ma$ $\vec{F} = \frac{d\vec{p}}{dt}$

N3 for every action there is an
equal opposite reaction

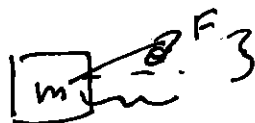
apply Force F to a mass m



or at an angle

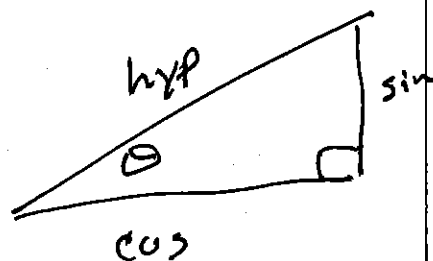


$$F_x = F \cos \theta$$



$$F_x = F \cos \theta$$

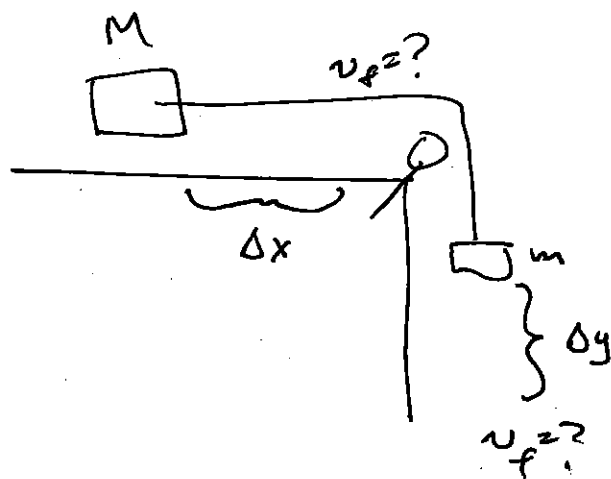
$$F_y = F \sin \theta$$



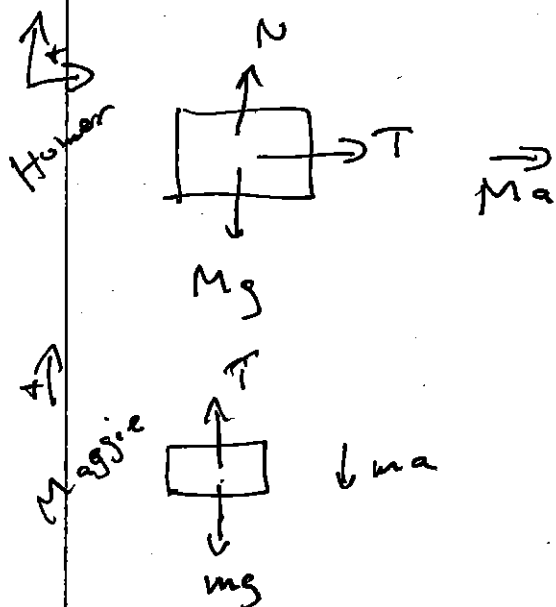
Homer and Maggie

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Homer Simpson $M = 150 \text{ kg}$ rests on the frictionless cliff edge above Springfield Gorge. He dangles ~~at~~ Maggie $m = 10 \text{ kg}$ over the edge. The rope supporting her is massless and frictionless as is the pulley,



When Homer slides $\Delta x = 5 \text{ m}$ to the cliff edge, how fast is he sliding at the edge? $v_i = 0$



$$\Sigma F_y: N - Mg = 0 \rightarrow N = Mg$$

$$\Sigma F_x: T = Ma$$

$$\Sigma F_y: T - mg = -ma$$

$$Ma - mg = -ma$$

$$(m + M)a = mg$$

$$a = \frac{m}{M+m} g = \frac{10 \text{ kg}}{160 \text{ kg}} 9.8 \text{ m/s}^2$$

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$$a = 0.6125 \text{ m/s}^2 \rightarrow \text{for Homer} \quad \downarrow \text{for Maggie}$$

a is the accel in the direction chosen,

$$v_f (\text{Homer}) = ?$$

$$v_i = 0$$

$$a = 0.6125 \text{ m/s}^2$$

$$x_i = 0$$

$$x_f = 5 \text{ m}$$

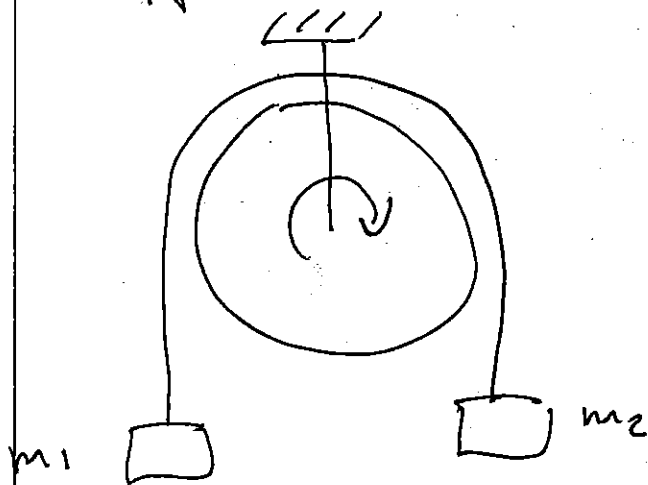
$$v_f^2 = v_i^2 + 2a(x_f - x_i)$$

$$v_f = \sqrt{2 \cdot 0.6125 \frac{\text{m}}{\text{s}^2} \cdot 5 \text{ m}}$$

$$v_f = 2.47 \text{ m/s} \rightarrow$$

$$v_f (\text{Maggie}) = 2.47 \text{ m/s} \downarrow$$

Atwood's Machine



$$m_1 < m_2$$

$$m_1 = 10 \text{ kg}$$

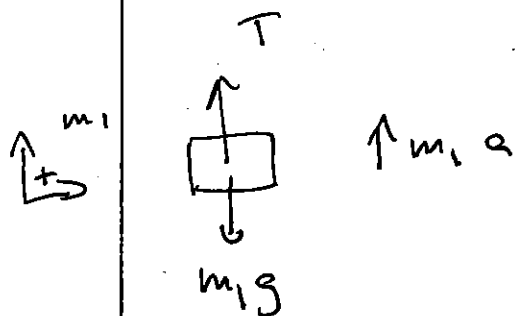
$$m_2 = 11 \text{ kg} \rightarrow \text{clockwise rotation}$$

pulley and cord are massless and frictionless

What is upward speed of m_1 when it has risen 2m?

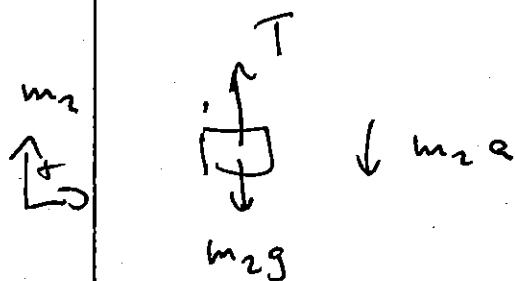
FREE BODY DIAGRAMS

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$$T - m_1 g = m_1 a$$

$$T = m_1 g + m_1 a$$



$$T - m_2 g = -m_2 a$$

eliminate T between the
2 equations

$$m_1 g + m_1 a - m_2 g = -m_2 a$$

$$(m_1 + m_2) a = (m_2 - m_1) g$$

$$a = \frac{m_2 - m_1}{m_1 + m_2} g = \frac{1 \text{ kg}}{21 \text{ kg}} \cdot 9.8 \frac{\text{m}}{\text{s}^2} = 0.47 \frac{\text{m}}{\text{s}^2}$$

$$v_f^2 = v_i^2 + 2a(y_f - y_i)$$

$$v_f = \sqrt{0 + 2 \cdot 0.47 \frac{\text{m}}{\text{s}^2} (2 \text{ m} - 0)}$$

$$= 1.37 \frac{\text{m}}{\text{s}} \quad \uparrow \text{ for } m_1 \quad \downarrow \text{ for } m_2$$

kg

Friction

$$F_f = \mu_k N \quad \text{while sliding}$$

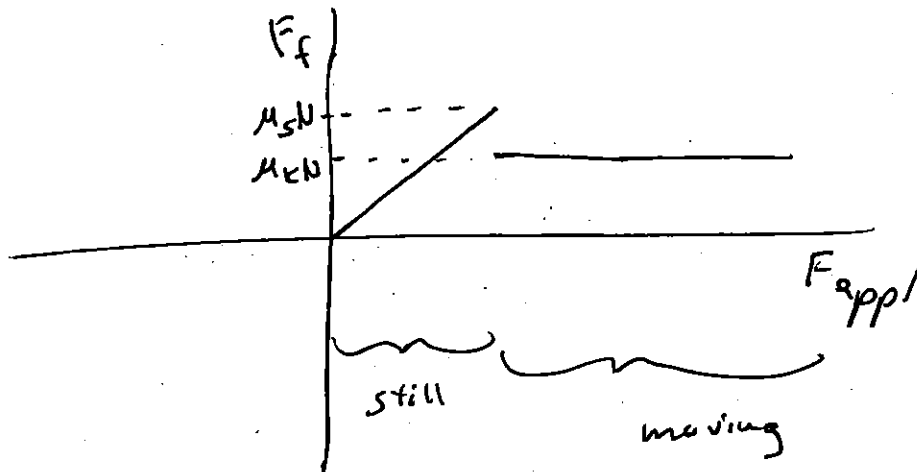
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in general $F_f \leq \mu N$

μ comes in moving μ_k kinetic

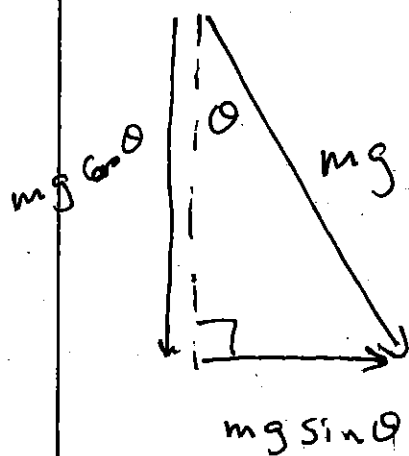
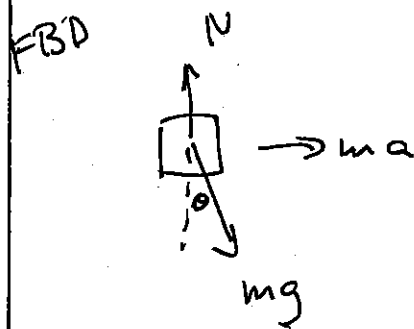
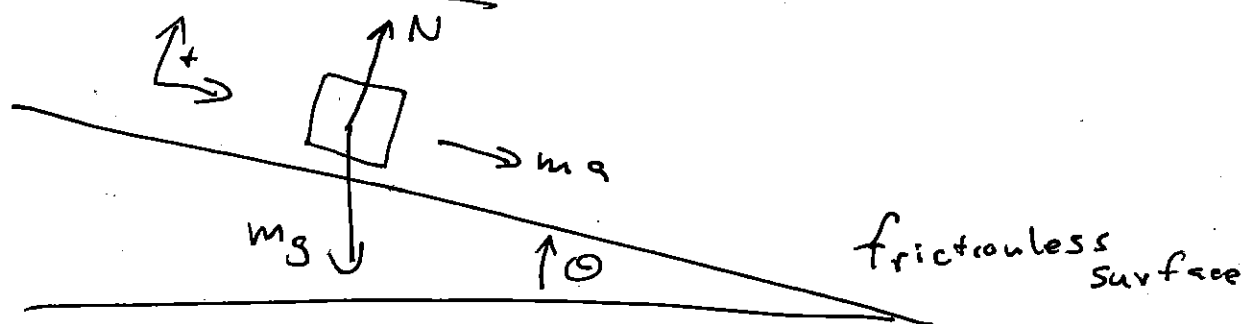
still μ_s static

$\mu_s > \mu_k$ in general



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Inclined Plane



$$\sum F_y: N - mg \cos \theta = 0$$

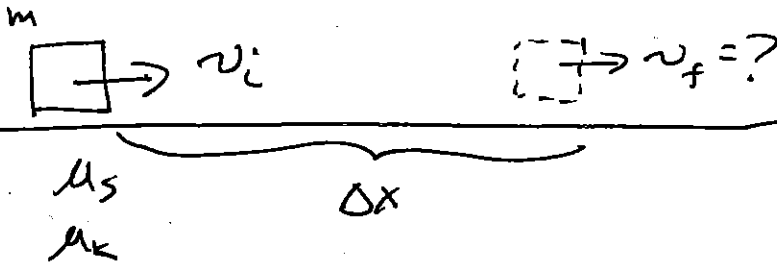
$$\boxed{N = mg \cos \theta}$$

$$\sum F_x: mg \sin \theta = ma$$

$$\boxed{a = g \sin \theta}$$

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Examples of motion w/ friction



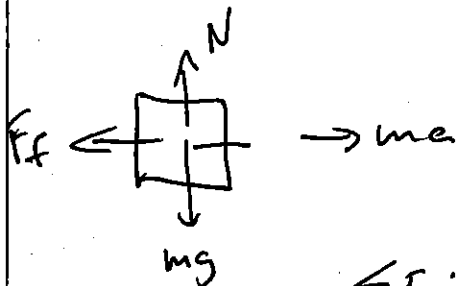
mass m is moving in $+x$ direction at $v_i = 3 \text{ m/s}$
 on surface $\mu_s = 0.4$, $\mu_k = 0.3$, slides $\Delta x = 2 \text{ m}$,
 How fast is it moving?

$$v_f^2 = v_i^2 + 2a(x_f - x_i)$$

$$x_i = 0 \quad x_f = 2 \text{ m}$$

$$v_i = 3 \text{ m/s} \quad v_f = ?$$

FBD to get a



$$F_f = \mu_k N \quad \text{since it's moving right}$$

$$\sum F_y: N - mg = 0 \quad N = mg$$

$$\sum F_x: -F_f = ma$$

$$-\mu_k mg = ma$$

$$\boxed{a = -\mu_k g}$$

$$v_f = \sqrt{v_i^2 + 2(-\mu_k g)x_f}$$

$$v_f = \sqrt{3^2 + -2 \cdot 0.3 \cdot 9.8 \cdot 2}$$

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$$= \sqrt{9 - 11.76} \quad \text{imaginary} \quad \text{no } v_f$$

(ie, it doesn't make it 2m!)

How far does it travel until $v_f = 0$?

$$v_f = 0$$

$$v_i = 3 \text{ m/s}$$

$$v_f^2 = v_i^2 + 2a(x_f - x_i)$$

$$a = -\mu_k g$$

$$0 = v_i^2 + 2ax_f$$

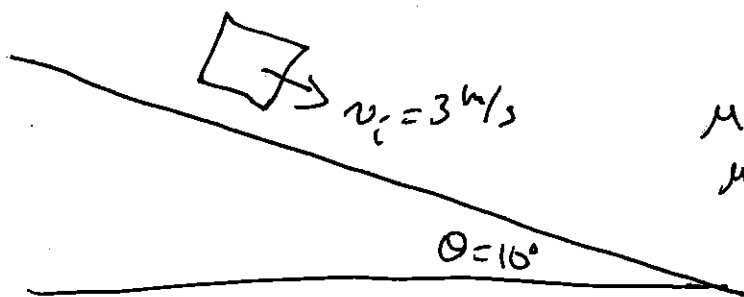
$$x_f = ? \quad x_i = 0$$

$$0 = v_i^2 - 2\mu_k g x_f$$

$$x_f = \frac{v_i^2}{2\mu_k g} = \frac{3^2}{2 \cdot 0.3 \cdot 9.8} = \boxed{1.53 \text{ m}}$$

$$m = 10 \text{ kg}$$

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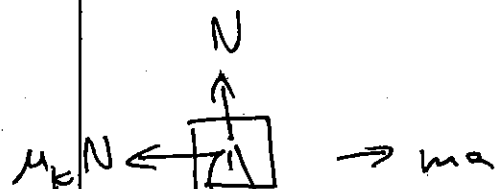


$$\mu_s = 0.3$$

$$\mu_k = 0.2$$

How fast is it moving after 3m?

$$F_f = \mu_k N \text{ (it's moving)}$$



$$\Sigma F_y: N - mg \cos \theta = 0 \quad N = mg \cos \theta$$

$$\Sigma F_x: mg \sin \theta - \mu_k N = ma$$

$$mg \sin \theta - mg \mu_k \cos \theta = ma$$

$$a = g (\sin \theta - \mu_k \cos \theta)$$

$$v_f^2 = v_i^2 + 2a(x_f - x_i)$$

$$v_f = \sqrt{v_i^2 + 2gx_f(\sin \theta - \mu_k \cos \theta)}$$

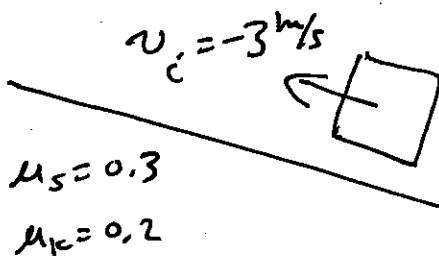
$$v_f = \sqrt{3^2 + 2 \cdot 9.8 \cdot 3 (\sin 10^\circ - 0.2 \cos 10^\circ)}$$

$$v_f = \sqrt{9 + 58.8 (0.174 - 0.197)}$$

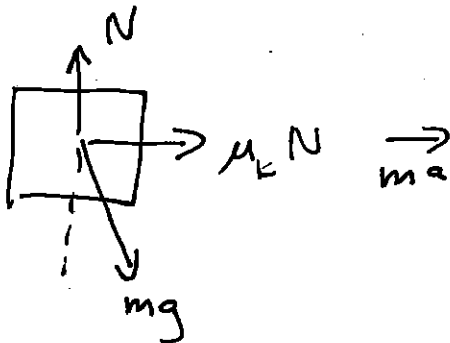
$$= \sqrt{9 + -1.35}$$

$$= 2.77 \text{ m/s}$$

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 $v_f = ?$ after 3m

$$x_f = -3 \text{ m} \quad x_i = 0 \text{ m}$$



$$N = mg \cos \theta$$

$$mg \sin \theta + \mu_k mg \cos \theta = ma$$

$$a = g (\sin \theta + \mu_k \cos \theta)$$

$$v_f = \sqrt{v_i^2 + 2a(x_f - x_i)}$$

$$= \sqrt{(-3)^2 + 2 \cdot 9.8 (0.174 + 0.2 \cdot 0.985) (-3)}$$

$$= \sqrt{9 + -58.8 \cdot 0.371}$$

$$= \sqrt{-12.8} \quad \text{it doesn't go 3m stops first}$$

$$x_f = ? \quad \cancel{v_f^2} = v_i^2 + 2g(\sin \theta + \mu_k \cos \theta)x_f$$

$$x_f = \frac{-v_i^2}{2g(\sin \theta + \mu_k \cos \theta)}$$

$$x_f = \frac{-(-3)^2}{2 \cdot 9.8 (0.371)} = \boxed{-1.24 \text{ m}}$$

$$m_1 = 100 \text{ kg}$$

$$v_i = 0$$

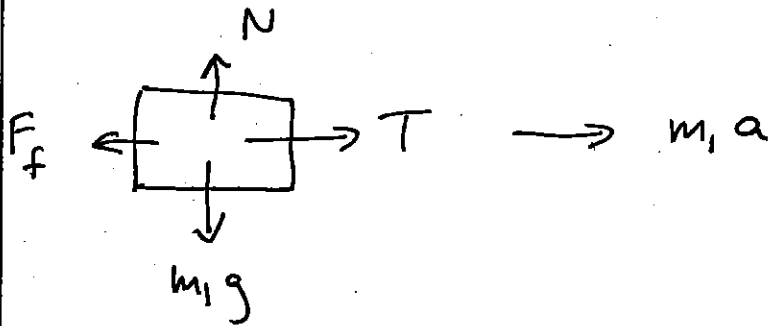
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$$\mu_s = 0.3$$

$$\mu_k = 0.2$$

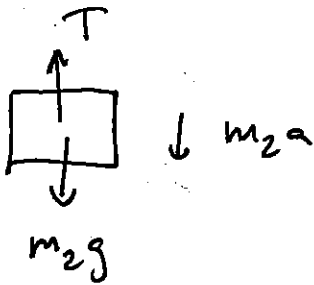
$$m_2 = 25 \text{ kg}$$

$$F_f = ?$$



$$N = m_1 g$$

$$T - F_f = m_1 a$$



$$T - m_2 g = -m_2 a$$

$$T = m_2 g - m_2 a$$

$$m_2 g - m_2 a - F_f = m_1 a$$

$$(m_1 + m_2) a = m_2 g - F_f$$

$$\text{if } a = 0$$

$$F_f = m_2 g$$

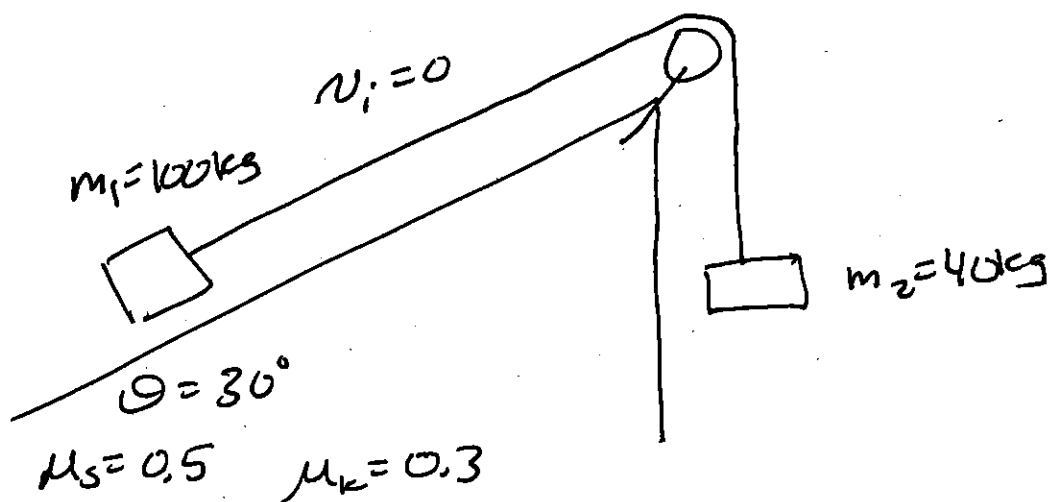
$$F_f = 245 \text{ N}$$

$$F_f \leq \mu_s m_1 g$$

$$F_f \leq 0.3 \cdot 100 \cdot 9.8$$

$$F_f \leq 294 \text{ N}$$

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What is F_f ? Does m_1 tend left or right?

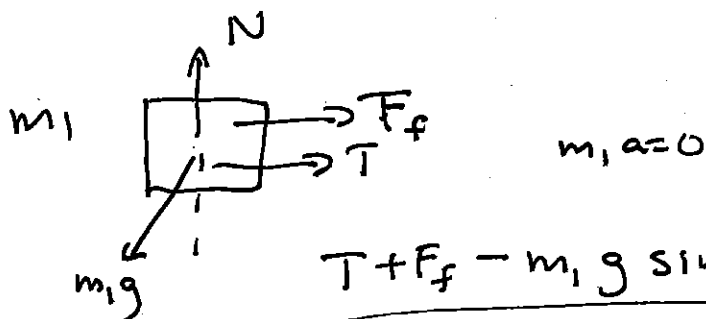
$$m_1 g \sin \theta = 50g \text{ to left}$$

$$T = m_2 g = 40g \text{ to right}$$

so m_1 is trying to

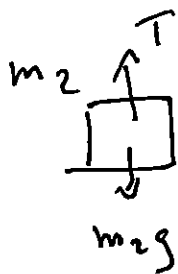
go left

$\therefore F_f$ to right!



$$T + F_f - m_1 g \sin \theta = 0$$

$$F_f = m_1 g \sin \theta - T$$



$$T = m_2 g$$

$$F_f = (m_1 \sin \theta - m_2) g$$

$$= (100 \cdot 0.5 - 40) 9.8$$

$$= 10 \cdot 9.8 = \boxed{98 \text{ N}}$$

$$F_{f \max} = \mu_s m_1 g \cos \theta = 0.5 \cdot 100 \cdot 9.8 \cdot 0.866$$

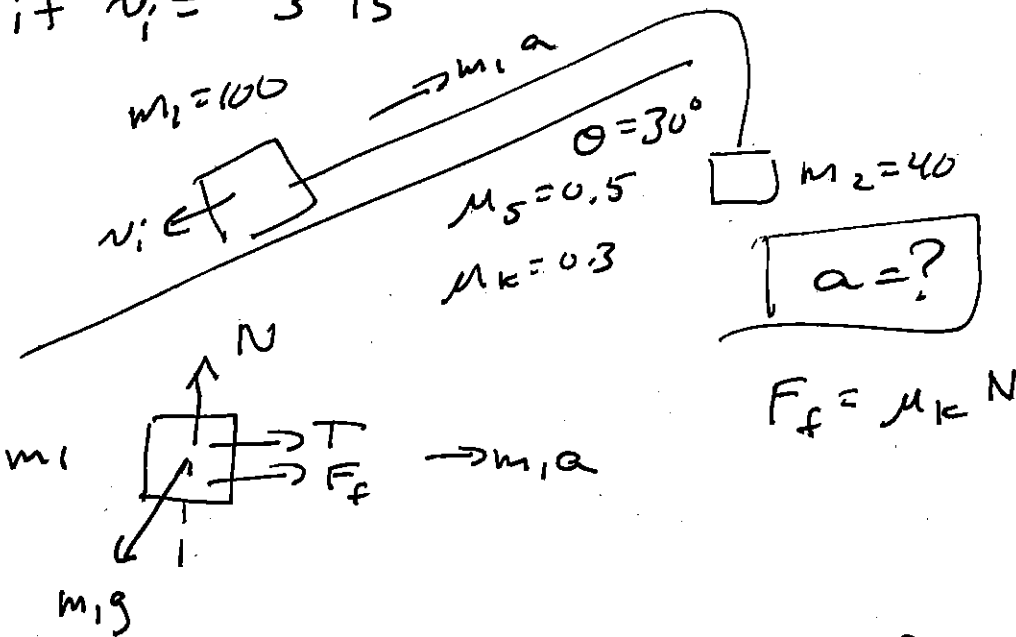
$$\leq 424 \text{ N}$$

98 N is available

m_1 has $a_f = 0$

if $v_i = -3 \text{ m/s}$

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$$F_f = \mu_k N$$

$$\sum F_y: N - m_1 g \cos \theta = 0 \quad N = m_1 g \cos \theta$$

$$\sum F_x: T + F_f - m_1 g \sin \theta = m_1 a$$

$$T + \mu_k m_1 g \cos \theta - m_1 g \sin \theta = m_1 a$$

$$m_2 \quad \begin{array}{c} \uparrow T \\ \downarrow m_2 g \end{array} \quad \downarrow m_2 a \quad T - m_2 g = -m_2 a$$

$$T = m_2 g - m_2 a$$

$$m_2 g - m_2 a + \mu_k m_1 g \cos \theta - m_1 g \sin \theta = m_1 a$$

$$a = \frac{g}{m_1 + m_2} (m_2 - m_1 \sin \theta + \mu_k m_1 \cos \theta)$$

$$a = \frac{9.8}{140} (40 - 100 \cdot 0.5 + 0.3 \cdot 100 \cdot 0.866)$$

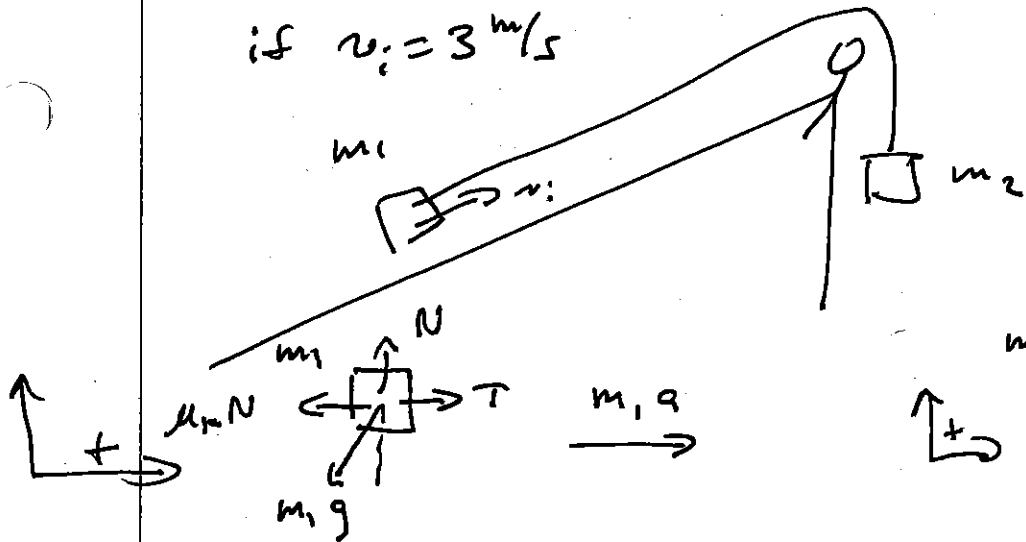
$$a = \frac{9.8}{140} (40 - 50 + 25.98)$$

> 0

$$a = 1.12 \text{ m/s}^2 \rightarrow$$

if $v_i = 3 \text{ m/s}$

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$$T - \mu_k N - m_1 g \sin \theta = m_1 a \quad T = m_2 g - m_2 a$$

$$m_1 a = m_2 g - m_2 a - \mu_k m_1 g \cos \theta - m_1 g \sin \theta$$

$$a = \frac{g}{m_1 + m_2} (m_2 - m_1 \sin \theta - \mu_k m_1 \cos \theta)$$

$$a = \frac{9.8}{140} (40 - 50 - 25.98)$$

$$= -2.52 \text{ m/s}^2 \quad \text{slows}$$